

```

1.import java.util.Scanner;
public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        // Input
        int num = sc.nextInt();
        // Calculate sum of digits
        int temp = num;
        int sum = 0;
        while (temp > 0) {
            sum += temp % 10;
            temp /= 10;
        }
        // Check Harshad condition
        if (num % sum == 0) {
            System.out.println("Harshad Number");
        } else {
            System.out.println("Not Harshad Number");
        }
        sc.close();
    }
}

```

```

2.import java.util.Scanner;
public class Solution {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();
        int sum = 0;
        // sum proper divisors (numbers < n that divide n)
        for (int i = 1; i <= n / 2; i++) {
            if (n % i == 0) {
                sum += i;
            }
        }
        if (sum > n) {
            System.out.println("Abundant Number");
        } else {
            System.out.println("Not Abundant Number");
        }
    }
}

```

```

3.import java.util.Scanner;

```

```

public class Solution {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();
        // Check if number has more than 2 digits
        if (n >= 100 || n <= -100) {
            int sum = 0;
            int num = Math.abs(n); // handle negative numbers too
            while (num > 0) {
                sum += num % 10;
                num = num / 10;
            }
            System.out.println("Sum of digit is " + sum);
        } else {
            System.out.println("Invalid Input");
        }
    }
}

```

```

4.import java.util.Scanner;
public class Solution {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt(); // start range
        int m = sc.nextInt(); // end range
        // Check constraints
        if (n < 1 || n > 20 || m < 1 || m > 20) {
            System.out.println("Invalid Input");
            return;
        }
        // Generate Fibonacci up to 20 terms
        int[] fib = new int[21];
        fib[1] = 0;
        fib[2] = 1;
        for (int i = 3; i <= 20; i++) {
            fib[i] = fib[i - 1] + fib[i - 2];
        }
        double sum = 0;
        for (int i = n; i <= m; i++) {
            sum += fib[i];
        }
        System.out.println("The Sum of Fibonacci value is " + sum);
    }
}

```

```

5.import java.util.Scanner;
public class Solution {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();
        if (n < 1 || n > 9) {
            System.out.println("Invalid Input");
        } else {
            for (int i = 1; i <= n; i++) {
                System.out.println(n + " x " + i + " = " + (n * i));
            }
        }
    }
}

```

```

6.import java.util.Scanner;
public class Solution {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();
        int m = sc.nextInt();
        // Check constraints
        if (n <= 0 || m <= 0 || n >= 30 || m >= 30 || n > m) {
            System.out.println("Invalid Input");
        } else {
            int sum = 0;
            for (int i = n; i <= m; i++) {
                if (i % 2 == 0) {
                    sum += i;
                }
            }
            System.out.println(sum);
        }
    }
}

```

```

7.import java.util.Scanner;
public class Solution {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();
        // Check if n is a 3-digit number
        if (n < 100 || n > 999) {

```

```

        System.out.println("No");
    } else {
        int temp = n;
        int sum = 0;
        while (temp > 0) {
            int digit = temp % 10;
            sum += digit * digit * digit; // cube of digit
            temp /= 10;
        }
        if (sum == n) {
            System.out.println("Yes");
        } else {
            System.out.println("No");
        }
    }
}
}

```

```

9. import java.util.*;
public class Solution {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        long n = sc.nextLong();
        // Constraint check
        if (n <= 0 || n > 1000000000) {
            System.out.println("Invalid Input");
            return;
        }
        long reversed = 0;
        while (n > 0) {
            long digit = n % 10;
            reversed = reversed * 10 + digit;
            n = n / 10;
        }
        System.out.println(reversed);
    }
}

```

```

8. import java.util.Scanner;
public class Solution {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        String s = sc.next().trim();
        if (s.length() == 1) {

```

```

        // single digit
        System.out.println("STDOUT");
    } else if (s.length() == 2) {
        // exactly two digits: swap them
        System.out.println("" + s.charAt(1) + s.charAt(0));
    } else {
        // more than two digits: take first two digits and swap them
        System.out.println("" + s.charAt(1) + s.charAt(0));
    }
}
}

```

```

10.import java.util.*;
public class Solution {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.println("Enter a number");
        int n = sc.nextInt();
        int count = 0;
        System.out.println(n); // print the starting number
        while (n != 1) {
            if (n % 2 == 0) {
                n = n / 2;
            } else {
                n = 3 * n + 1;
            }
            System.out.println(n);
            count++;
        }
        System.out.println("count:" + count);
    }
}

```

```

11.import java.util.*;
public class Solution {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        long num = sc.nextLong();
        // Check constraints
        if (num < 1 || num > 100000000) {
            System.out.println("Enter a Valid Input");
            return;
        }
        int count = 0;
    }
}

```

```

        long temp = num;
        while (temp > 0) {
            temp = temp / 10;
            count++;
        }
        System.out.println("The count of the given integer is: " + count);
    }
}

```

```

12.import java.util.Scanner;
public class Solution {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();
        if (n < 0 || n > 9) {
            System.out.println("Invalid Input");
            return;
        }
        char ch = 'A';
        for (int i = 1; i <= n; i++) {
            for (int j = 1; j <= i; j++) {
                System.out.print(ch + " ");
                ch++;
            }
            System.out.println();
        }
    }
}

```

```

13.import java.util.Scanner;

public class Solution {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();

        // Upper half including middle
        for (int i = 1; i <= n; i++) {
            // print spaces
            for (int s = 1; s <= n - i; s++) {
                System.out.print(" ");
            }
            // print letters
            char ch = 'A';

```

```

        for (int j = 1; j <= i; j++) {
            System.out.print(ch + " ");
            ch++;
        }
        System.out.println();
    }

    // Lower half
    for (int i = n - 1; i >= 1; i--) {
        // print spaces
        for (int s = 1; s <= n - i; s++) {
            System.out.print(" ");
        }
        // print letters
        char ch = 'A';
        for (int j = 1; j <= i; j++) {
            System.out.print(ch + " ");
            ch++;
        }
        System.out.println();
    }
}

14.import java.util.*;
public class Solution {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();
        for (int i = 0; i < n; i++) {
            for (int j = 0; j < n; j++) {
                // Print * for border rows and columns
                if (i == 0 || i == n - 1 || j == 0 || j == n - 1) {
                    System.out.print("*");
                } else {
                    System.out.print(" ");
                }
            }
            System.out.println(); // next line after each row
        }
    }
}

15.import java.util.*;

```

```
public class Solution {  
    public static void main(String[] args) {  
        Scanner sc = new Scanner(System.in);  
        int n = sc.nextInt();  
        for (int i = 0; i < n; i++) {  
            char ch = (char) ('A' + i); // starting character of each row  
            for (int j = 0; j <= i; j++) {  
                System.out.print((char)(ch + j) + " ");  
            }  
            System.out.println();  
        }  
    }  
}
```